PHASE 1 DOCUMENT

ComicCraft: AI Comic Story Creator using Gemini Models

Phase Name: Project Proposal and Requirement Analysis

Project Name	ComicCraft: AI Comic Story Creator using Gemini Models
Phase	Phase 1 – Proposal and Requirements
Project Type	Generative AI Comic Creation Application
Main Technology	Google Gemini Models
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 1 of **ComicCraft: AI Comic Story Creator using Gemini Models**. The purpose of this phase is to define the project concept, identify the storytelling problem it addresses, establish the objectives and scope, identify target users, and document the requirements that will guide design and development in later phases.

2. Project Abstract

ComicCraft is a generative-AI application that helps users create original comic stories from simple prompts. A user can describe a genre, characters, setting, mood, theme, or story idea, and ComicCraft uses Gemini models to generate a structured story concept, scene sequence, dialogue suggestions, narration, and panel-by-panel comic script. The application is designed to make comic storytelling more accessible to students, beginners, writers, and content creators while preserving user creativity and responsible AI use.

3. Problem Identification

Creating a comic requires several connected skills: developing a story, designing characters, arranging scenes, writing dialogue, pacing a narrative, and planning visual panels. Beginners may find it difficult to convert an idea into a coherent comic script, while experienced creators may spend significant time on brainstorming and first drafts. Existing writing tools may not provide guided support for transforming a short idea into a comic-specific structure. A tool is needed to help users plan comic narratives quickly without replacing their creative decisions.

4. Proposed Solution

ComicCraft will provide a user-friendly web application where a creator enters a prompt, selects options such as genre, tone, target audience, comic length, and style, and receives AI-assisted story content. Gemini models will be accessed through a secure backend service to generate a story outline, character descriptions, scenes, panel instructions, dialogue, and captions. Users can review, edit, regenerate, save, and export their comic scripts. The project focuses on text-based comic creation; visual image generation can be treated as an optional future enhancement.

5. Project Objectives

The objectives are to help users transform a basic idea into a comic story; generate structured story outlines and panel sequences; create character profiles, dialogues, narration, and scene descriptions; allow users to edit and customize AI-generated content; support multiple comic genres and tones; provide a simple and accessible creation workflow; protect Gemini API credentials through secure backend integration; and promote original, ethical, and responsible content creation.

6. Scope of Phase 1

Phase 1 covers proposal and requirement analysis for ComicCraft. It documents the project purpose, users, features, requirements, boundaries, dependencies, and success criteria. This phase does not include interface coding, database implementation, Gemini API programming, image generation, deployment, or final testing. These activities will be addressed in subsequent phases.

7. Target Users

Primary users include students working on creative assignments, beginner comic writers, storytellers, content creators, and hobbyists. Educators may use the platform as a guided storytelling tool in appropriate learning activities. Administrators manage user access, content categories, system configuration, and application policies. The first version can prioritize individual creators and later expand to collaboration or classroom features.

8. Functional Requirements

The system shall allow users to register and log in. The system shall allow a user to enter a comic idea or prompt. The system shall allow selection of story settings such as genre, tone, audience, comic length, and preferred language where supported. The system shall use Gemini models to generate a story title, premise, outline, characters, scenes, dialogue, captions, and panel-by-panel script. The system shall allow users to edit, regenerate, save, and delete their own comic drafts. The system shall organize saved comic projects in a user dashboard. The system shall provide suitable messages when generation is unavailable or input is incomplete.

9. Non-Functional Requirements

ComicCraft should provide a responsive and easy-to-use interface on mobile and desktop devices. User accounts and saved comics should be protected through authentication and authorization. Gemini API keys must be stored only in secure backend configuration and must never be exposed in the frontend. The system should provide reasonable response times, understandable loading and error states, scalable service design, and privacy-aware handling of user prompts and saved content.

10. Proposed Comic Creation Flow

1. User logs in to ComicCraft.
2. User selects Create New Comic.
3. User enters a story idea and selects desired settings.
4. The application validates the input.
5. The backend creates a structured prompt and sends it to Gemini.
6. Gemini returns a comic outline and panel-wise script.
7. ComicCraft displays the title, characters, scenes, dialogue, captions, and panel descriptions.
8. User edits, regenerates, saves, or exports the comic script.
9. The user can return later to continue working on the saved project.

11. System Boundaries and Ethics

ComicCraft is a creative-assistance tool and should not claim ownership of a user’s original concept. Users remain responsible for reviewing generated material, respecting copyright, avoiding plagiarism, and following institutional or platform policies. The system should discourage hateful, harmful, explicit, deceptive, or unlawful content. It should not imitate living artists or copyrighted characters in a misleading way. Generated output should be treated as a draft that the user improves through their own creative judgement.

12. Assumptions and Dependencies

The project assumes that users have an internet connection and a supported browser. It depends on access to Gemini models through an authorized API, a secure backend environment, database or cloud storage for user projects, and a frontend interface. Output quality depends on the user prompt, selected settings, model availability, and system-level content safety rules.

13. Success Criteria

Phase 1 will be considered complete when the project problem, objectives, scope, users, functional requirements, non-functional requirements, content boundaries, dependencies, workflow, and future modules are clearly documented. The full project will be successful when a user can create, edit, save, and retrieve an original panel-by-panel comic script from a simple idea using a secure and user-friendly AI-assisted workflow.

14. Planned Next Phase

The next phase will be **Phase 2: System Design and Architecture**. It will define the ComicCraft application architecture, frontend and backend modules, data model, user interface flow, comic-generation pipeline, user roles, and secure Gemini model integration design.

15. Conclusion

Phase 1 establishes the foundation for ComicCraft. The project proposes a practical AI-assisted system that makes comic storytelling easier to begin and structure while retaining the creator's control over the final work. The documented requirements will guide design, implementation, testing, and deployment in the remaining project phases.

Phase Approval

Prepared by	_____
Guide / Faculty	_____
Approved by	_____